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Project Title:	Effects of Simulated Microgravity on Sumoylation		
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Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
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Key Personnel Changes/Previous PI:	no changes in key personnel.		
COI Name (Institution):	Hartson, Steven Ph.D. (Oklahoma State University)		
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Task Description:	Many of the adaptive responses to stress that a cell should need during spaceflight are regulated on Earth by Small Ubiquitin-like Modifier (SUMO). SUMO is a small ubiquitin-like protein that regulates a large number of basic and fundamental cellular processes. SUMO can be attached to lysine residues within target proteins as a post-translational modification, altering their function and cellular localization. SUMO can also bind non-covalently to other proteins through their SUMO-Interacting Motifs (SIMs). SUMO mediates cellular responses to a large number of stresses including hypoxia, radiation, thermo-tolerance, and oxidative stress. However, a significant gap in knowledge exists about how the SUMO signal transduction system is used by the cell to sense and regulate its response to stresses found in space, and microgravity in particular. The long-term goal of the Miller lab is to understand how signal transduction circuits control microtubules so that effective and targeted interventions can be developed to manipulate microtubule functions in stressful situations like disease and space travel. The Miller lab recently established that the microtubule-associated protein Stu2p interacts with SUMO. Preliminary data from the Miller lab suggest that there are two modes of interaction between Stu2p and SUMO: a covalent interaction in which Stu2p is conjugated by SUMO and a non-covalent interaction in which Stu2p simply binds to SUMO. The objective of this work is to identify novel signal transduction mechanisms that regulate cytoskeletal networks in response to simulated microgravity conditions. The rationale is that by achieving a fuller understanding of the regulation of cytoskeletal polymers under simulated microgravity conditions on Earth, researchers can develop more complete hypotheses of how the cytoskeleton responds to gravitational changes in the space environment, which can be tested in future space missions. This knowledge may be used to design interventions for adverse health effects that are associated with space travel. The specific goal of this proposal is to determine how simulated microgravity alters the interaction of SUMO with the cellular proteome, and microtubule-associated proteins (MAPS) in particular. The central hypothesis of this work is that microgravity alters the post-translational modifications of lysine residues of a wide variety of cytoskeletal proteins. To test this hypothesis, the post-translational modifications on lysines residues will be analyzed on a proteome-wide basis by mass-spectrometry and immunoblotting under simulated microgravity conditions. We will also test whether the proteome binds differentially to SUMO when it experiences microgravity. An additional emphasis will be placed on the MAP Stu2p. This work is significant because it will help identify sumoylation as a novel molecular mechanism at the biochemical and molecular level within the yeast cell that senses and responds to the environmental factors of space. Understanding these effects on basic cell biology may lead to a better understanding of how space travel affects normal tissue function, immune function, stem cell function, tissue regeneration, and embryogenesis in humans. As tubulin and the human version of Stu2 play important roles in cell division, these studies may also shed light on molecular mechanisms that are important in cancer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Impact of this research is that it will provide new insights into how cells divide and the molecular mechanisms that they use to mitigate various cellular stresses.
Task Progress:	The Small Ubiquitin-like Modifier (SUMO), is a protein that is an important regulator of several of the cell's responses to stress. SUMO functions in two ways. It is a post-translational modification that attaches covalently to lysine amino acids within a target protein, altering its function. SUMO can also bind to other proteins non-covalently, through their SUMO-Interacting Motifs, known as SIMs. The mechanisms by which SUMO regulates cellular stress under earth's normal gravity are only partially understood. Even less is understood about how, and/or the extent to which, SUMO regulates cellular stress responses in space or simulated-microgravity. Other laboratories have previously observed dysfunction of the cytoskeleton under microgravity conditions (Najrana, et al. 2016; Purevdorj-Gage, et al. 2006, and Masiello, Maria Grazia, et al. 2014). [Ed. Note: See References below.] We hypothesize that this could be mediated by either one or both types of SUMO interactions. This hypothesis makes the prediction that SUMO levels should change under simulated microgravity conditions. To test this, we determined how simulated microgravity would alter both the covalent and non-covalent interactions of SUMO with the cellular proteome. To simulate microgravity, we grew the baker's yeast Saccharomyces cerevisiae in a rotating wall vessel (RWV), which significantly reduces the acceleration forces that yeast experience due to the time they are in free fall. In Aim 1, we asked, "To what extent does simulated microgravity alter the sumoylation of proteins within the cell?" For this analysis, we used a labeling technique called, "Stable Isotope Labeling by/with Amino acids in Cell culture" (SILAC) to differentiate between the proteins of cells grown in normal gravity and the cells grown in simulated microgravity. We used mass spectrometry to identify the lysine residues to which SUMO attaches to its target proteins. In this experiment, we identified 268 peptides that contained at least one SUMO stub, corresponding to 253 different proteins. Of these, 54 proteins have been previously published as SUMOylated. Thus, this analysis identified the modified lysine for 199 new SUMOylated proteins. To determine whether simulated microgravity (SMG) induced changes in the abundance of these SUMOylated proteins, we analyzed the differences in the light/heavy SILAC ratios of the sumoylated proteins. This identified 38 proteins for which abundances were altered reproducibly across 6 bio-reps at the 95% confidence level. Of these, 12 proteins were decreased (>50%) in SMG, 6 were increased (>50%) in SMG, and 20 changed less than 50%. Notably, the sumoylated peptide from one microtubule-associated protein (MAP) was decreased in simulated microgravity, but the same non-sumoylated MAP peptide was increased in microgravity. This suggests the possibility that simulated microgravity is regulating the stability of some MAPs via sumoylation. In this model, SUMO is causing the degradation of the MAP in microgravity. We speculate that this may be one of the molecular mechanisms by which microgravity alters mitosis. In Aim 2, we asked "To what extent does simulated microgravity alter the non-covalent binding of cellular proteins to SUMO?" Non-covalent SUMO interactions are important for regulating many cellular functions. Unlike the single attachment point of a covalent SUMOylation at its C-terminal glycine, non-covalent interactions of SUMO can occur on three distinct binding faces of the SUMO molecule. These SUMO binding faces can interact with other proteins that contain a SUMO Interacting Motif (SIM). In order to uncover novel SUMO interacting proteins in a way that could provide insight into the binding faces, we used three different versions of SUMO that cannot be conjugated to target proteins. Yeast cells were SILAC-labeled and treated with simulated microgravity by being grown in a horizontally positioned RWV. Whole extracts were prepared from each of these conditions and the binding of cellular proteins to

non-covalent forms of SUMO was analyzed by mass spectrometry for each condition.

For this experiment, we first determined the total pool of proteins that could bind non-covalently to the three different SUMO columns. This identified over 1,500 proteins that could bind to one or more versions of SUMO, rediscovering 57% of the published SUMO interactome. This also confirmed 34% of our SUMO interactome and identified 1004 novel SUMO-interacting proteins. Significantly, 289 proteins bound to at least two of the three versions of SUMO. 122 proteins bound to all three versions of SUMO. Since this experiment was conducted in non-denaturing conditions, the proteins identified include those that bind directly to SUMO and indirectly to SUMO through a complex with a direct SUMO-interacting protein.

To determine whether simulated microgravity induced changes in the abundance of the proteins bound to SUMO, we analyzed the differences in the light/heavy SILAC ratios of the SUMO-bound proteins. The noncovalent SUMO columns enriched 18 proteins whose abundance was changed in one condition relative to the other. Three of these proteins displayed a decreased protein abundance in simulated microgravity, while the other 15 proteins displayed an increased abundance in simulated microgravity. It is notable that the binding of alpha-tubulin was enriched in simulated microgravity, binding to two of the three SUMO columns. Unexpectedly, unlike the covalent SUMOylation experiment, which displayed a greater number of proteins decreased in simulated microgravity, the majority of proteins enriched by the noncovalent SUMO columns displayed increased protein abundance in simulated microgravity. Despite this difference, many of the same cellular functions as seen in the SUMOylation profile were also enriched by SUMO noncovalently. This includes a cytoskeletal protein, cell wall components, DNA damage repair proteins, and ribosome components.

Together, these results support our model that simulated microgravity alters the covalent and non-covalent interaction of key cellular proteins with SUMO.

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Masiello, Maria Grazia, et al. "Phenotypic switch induced by simulated microgravity on MDA-MB-231 breast cancer cells." BioMed research international 2014 (2014).

Bibliography Type:	Description: (Last Updated: 01/19/2026)
Articles in Peer-reviewed Journals	Sabo J, Miller R. "Long-term effects of simulated microgravity on protein SUMOylaiton and non-covalent SUMO interactions." Journal of Biological Chemistry. 2023 Jan 1;299(3):S377-8. https://doi.org/10.1016/j.jbc.2023.103718 , Jan-2023
Articles in Peer-reviewed Journals	Sabo JA, Hartson SD. "Simulated microgravity-induced changes in SUMOylation and protein expression in Saccharomyces cerevisiae." Int J Mol Sci. 2025 Dec 19;27(1):42. https://doi.org/10.3390/ijms27010042 ; PMID: 41515919; PMCID: PMC12786128 , Dec-2025
Dissertations and Theses	Jurko D. "Determination of the quantity of Earth's gravity imparted on particles in a rotating cylinder fluid regime based on particle density and rotation speed. " Master's Thesis, Oklahoma State University, March 2023. , Mar-2023